

ECE 332 — Homework 1: Generated Answers from Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

Problem 1 — Current Direction in Changing B Field

15 pts

Setup: Loop in x - y plane, $ds = \hat{z} da$. $\mathbf{B} = \hat{z}B_0 \sin(\omega t)$.

EMF & FARADAY'S LAW									
Transformer EMF <i>stationary loop, B changes</i>									
$V_{\text{emf}} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$									
Motional EMF <i>loop moves, B static</i>									
$V_{\text{emf}} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$									
General (flux form) <i>always works</i>									
$V_{\text{emf}} = - \frac{d\Phi}{dt} \text{ (V)} \quad \Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \text{ (V}\cdot\text{s)}$									
N-turn coil <i>transformer, N turns</i>									
$V_{\text{emf}} = -N \frac{d\Phi}{dt} = - \frac{d\lambda}{dt} \text{ (V)} \quad \lambda = N\Phi \text{ (V}\cdot\text{s)}$									
Uniform B shortcut <i>B same everywhere on loop</i>									
	<table> <tr> <th>$B(t)$</th><th>$\partial B / \partial t$</th></tr> <tr> <td>$B_0 \sin(\omega t)$</td><td>$B_0 \omega \cos(\omega t)$</td></tr> <tr> <td>$B_0 \cos(\omega t)$</td><td>$-B_0 \omega \sin(\omega t)$</td></tr> <tr> <td>$B_0 e^{-\alpha t}$</td><td>$-\alpha B_0 e^{-\alpha t}$</td></tr> </table>	$B(t)$	$\partial B / \partial t$	$B_0 \sin(\omega t)$	$B_0 \omega \cos(\omega t)$	$B_0 \cos(\omega t)$	$-B_0 \omega \sin(\omega t)$	$B_0 e^{-\alpha t}$	$-\alpha B_0 e^{-\alpha t}$
$B(t)$	$\partial B / \partial t$								
$B_0 \sin(\omega t)$	$B_0 \omega \cos(\omega t)$								
$B_0 \cos(\omega t)$	$-B_0 \omega \sin(\omega t)$								
$B_0 e^{-\alpha t}$	$-\alpha B_0 e^{-\alpha t}$								

Find V_{emf} first:

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = B_0 \sin(\omega t) \cdot A \quad V_{\text{emf}} = - \frac{d\Phi}{dt} = -AB_0\omega \cos(\omega t)$$

From the SIGN table: $V_{\text{emf}} < 0 \Rightarrow$ current $-\hat{\phi}$ (CW); $V_{\text{emf}} > 0 \Rightarrow +\hat{\phi}$ (CCW); $V_{\text{emf}} = 0 \Rightarrow$ no current.

(a) $t = 0$: $V_{\text{emf}} = -AB_0\omega \cos(0) = -AB_0\omega < 0$

I flows in $-\hat{\phi}$ (clockwise)
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(b) $\omega t = \pi/4$: $\cos(\pi/4) = \sqrt{2}/2 > 0 \Rightarrow V_{\text{emf}} < 0$

I flows in $-\hat{\phi}$ (clockwise)
--

(c) $\omega t = \pi/2$: $\cos(\pi/2) = 0 \Rightarrow V_{\text{emf}} = 0$

$I = 0$ — no current at this instant

Explanation: At $t = 0$ and $\omega t = \pi/4$, B is increasing (positive and growing), so by Lenz's law the induced current opposes the increase by creating flux in $-\hat{z}$, requiring CW ($-\hat{\phi}$) current. At $\omega t = \pi/2$, $B = B_0$ is at its peak and instantaneously not changing, so $V_{\text{emf}} = 0$.

Problem 2 — Square Loop Near Long Wire

14 pts

Setup: Wire along z -axis; square loop in y - z plane, $a = 5$ cm, $b = 15$ cm, height $c = 10$ cm.
 $I(t) = 5 \cos(2\pi \times 10^4 t)$ A. Total resistance: $R_{\text{tot}} = 1 + 4 = 5 \Omega$.

EMF & FARADAY'S LAW	
Transformer EMF <i>stationary loop, B changes</i> $V_{\text{emf}} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$	$\mathbf{B}_0 \text{ (const)} \quad 0$
Motional EMF <i>loop moves, B static</i> $V_{\text{emf}} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$	
General (flux form) <i>always works</i> $V_{\text{emf}} = - \frac{d\Phi}{dt} \text{ (V)} \quad \Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \text{ (V}\cdot\text{s)}$	
N-turn coil <i>transformer, N turns</i> $V_{\text{emf}} = -N \frac{d\Phi}{dt} = - \frac{d\lambda}{dt} \text{ (V)} \quad \lambda = N\Phi \text{ (V}\cdot\text{s)}$	
Uniform B shortcut <i>B same everywhere on loop</i>	
B FIELDS	
Infinite wire <i>single wire carrying I, N = 1</i> $\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r}$	$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r} \quad \mu = \mu_r \mu_0$
Inside toroid <i>coil of N turns carrying I</i> $\mathbf{B} = \hat{\phi} \frac{\mu_0 N I}{2\pi r}$	<i>N in B only if the source (current-carrying thing) is a multi-turn coil. Single wire $\Rightarrow N = 1$, drop it. N in $V_{\text{emf}} = -Nd\Phi/dt$ is the receiver coil turns.</i>
$\partial B / \partial t$ TABLE	

(a) Find V_{emf} :

Using the long wire + rect loop formula from the cheat sheet:

$$V_{\text{emf}} = - \int_0^c \int_a^b \frac{\partial}{\partial t} \left(\frac{\mu_0 I}{2\pi r} \right) dr dz = - \frac{\mu_0 c}{2\pi} \frac{dI}{dt} \ln \frac{b}{a}$$

$$\frac{dI}{dt} = -5(2\pi \times 10^4) \sin(2\pi \times 10^4 t) = -\pi \times 10^5 \sin(2\pi \times 10^4 t)$$

$$V_{\text{emf}} = - \frac{(4\pi \times 10^{-7})(0.10)}{2\pi} \cdot (-\pi \times 10^5) \cdot \ln \frac{0.15}{0.05} \cdot \sin(2\pi \times 10^4 t)$$

$$= \frac{(4\pi \times 10^{-7})(0.10)}{2\pi} \cdot \pi \times 10^5 \cdot \ln 3 \cdot \sin(2\pi \times 10^4 t)$$

$$= (2 \times 10^{-8}) \cdot (\pi \times 10^5) \cdot (1.099) \cdot \sin(2\pi \times 10^4 t)$$

$$V_{\text{emf}} \approx 6.9 \times 10^{-3} \sin(2\pi \times 10^4 t) \text{ V} = 6.9 \text{ mV} \cdot \sin(2\pi \times 10^4 t)$$

(b) $I_{\text{loop}} = V_{\text{emf}} / (R_{\text{int}} + R_{\text{ext}}) = V_{\text{emf}} / 5$

$$I_{\text{loop}} = \frac{6.9 \times 10^{-3}}{5} \sin(2\pi \times 10^4 t) \approx 1.38 \text{ mA} \cdot \sin(2\pi \times 10^4 t)$$

Peak magnitude: **1.38 mA**. Direction: when $V_{\text{emf}} > 0$ (positive half-cycle), current flows in $+\hat{\phi}$ in the loop (CCW when viewed from $+x$).

Problem 3 — Toroid Transformer

15 pts

Setup: Wire along z : $I = I_0 \cos \omega t$. Iron toroid (μ_r), $N = 100$ turns, inner radius a , outer radius b , height c .

EMF & FARADAY'S LAW	
Transformer EMF <i>stationary loop, B changes</i>	$V_{\text{emf}} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$
Motional EMF <i>loop moves, B static</i>	$V_{\text{emf}} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$
General (flux form) <i>always works</i>	$V_{\text{emf}} = - \frac{d\Phi}{dt} \text{ (V)} \quad \Phi = \int_S \mathbf{B} \cdot d\mathbf{s} \text{ (V}\cdot\text{s)}$
N-turn coil <i>transformer, N turns</i>	$V_{\text{emf}} = -N \frac{d\Phi}{dt} = - \frac{d\lambda}{dt} \text{ (V)} \quad \lambda = N\Phi \text{ (V}\cdot\text{s)}$
Uniform B shortcut <i>B same everywhere on loop</i>	$B_0 \text{ (const)} \quad \quad \quad 0$
SIGN → CURRENT DIRECTION	
$B = \hat{\phi} \frac{\mu_0 I}{2\pi r}$	

(a) Derive V_{emf} :

$\mathbf{B}$ inside toroid from wire (source has 1 turn, so $N_{\text{source}} = 1$): $\mathbf{B} = \hat{\phi} \frac{\mu_r \mu_0 I}{2\pi r}$

Flux through one turn of the 100-turn coil (surface normal $\hat{\phi}$, integrating in r - z plane):

$$\Phi_1 = \int_0^c \int_a^b \frac{\mu_r \mu_0 I_0 \cos \omega t}{2\pi r} dr dz = \frac{\mu_r \mu_0 c}{2\pi} I_0 \cos \omega t \cdot \ln \frac{b}{a}$$

$$V_{\text{emf}} = -N \frac{d\Phi_1}{dt} = \frac{N \mu_r \mu_0 c}{2\pi} I_0 \omega \sin \omega t \cdot \ln \frac{b}{a}$$

$$V_{\text{emf}} = \frac{N \mu_r \mu_0 c I_0 \omega}{2\pi} \ln \frac{b}{a} \cdot \sin \omega t \text{ (V)}$$

(b) Evaluate $f = 60$ Hz, $\mu_r = 4000$, $a = 5$ cm, $b = 6$ cm, $c = 2$ cm, $I_0 = 50$ A:

$$\omega = 2\pi \times 60 = 120\pi \text{ rad/s}; \quad \ln(6/5) = \ln 1.2 = 0.182$$

$$\frac{N \mu_r \mu_0 c}{2\pi} = \frac{100 \times 4000 \times 4\pi \times 10^{-7} \times 0.02}{2\pi} = \frac{100 \times 4000 \times 4 \times 10^{-7} \times 0.02}{2} = 16 \times 10^{-3}$$

$$V_{\text{emf}} = 16 \times 10^{-3} \times 50 \times 120\pi \times 0.182 \cdot \sin \omega t = 16 \times 10^{-3} \times 50 \times 68.68 \cdot \sin \omega t$$

$$|V_{\text{emf}}|_{\text{peak}} \approx 54.9 \text{ mV} \cdot \sin(120\pi t) \text{ V}$$

Wait — recalculating:

$$16 \times 10^{-3} \times 50 = 0.8; \quad 0.8 \times 120\pi \times 0.182 = 0.8 \times 68.68 = 54.9 \text{ V peak? Let me recheck.}$$

$$= 100 \times \frac{4\pi \times 10^{-7} \times 4000 \times 0.02}{2\pi} \times 50 \times 120\pi \times 0.182 = 100 \times 16 \times 10^{-6} \times 50 \times 120\pi \times 0.182 \approx \boxed{5.50 \text{ V (peak)}}$$

Problem 4 — Electromagnetic Generator

6 pts

Setup: $A = 0.1 \text{ m}^2$, $B_0 = 0.4 \text{ T}$, 3600 rpm, $R = 150 \Omega$.

emf	inside source	term.
< 0	$- \rightarrow +$ terminal	$-\hat{\phi}$ (CW)
> 0	$+ \rightarrow -$ terminal	$+\hat{\phi}$ (CCW)
$= 0$	—	none

Lenz: induced I always opposes the change in Φ .

ln INTEGRAL

$B \propto 1/r$, loop spans $r = a$ to $r = b$
 $\int_a^b dr$

$$\omega = 2\pi f = 2\pi \times \frac{3600}{60} = 2\pi \times 60 = 120\pi \text{ rad/s}$$

From the cheat sheet ROTATING LOOP:

$$I_{\text{amp}} = \frac{AB_0\omega}{R} = \frac{0.1 \times 0.4 \times 120\pi}{150} = \frac{4.8\pi}{150} = \frac{15.08}{150}$$

$$I_{\text{amp}} \approx 0.1005 \text{ A} \approx 100.5 \text{ mA}$$

Problem 5 — Displacement Current (Capacitor)

6 pts

Setup: $\varepsilon = 4\varepsilon_0$, $A = 10 \text{ cm}^2 = 10^{-3} \text{ m}^2$, $d = 2 \text{ cm} = 0.02 \text{ m}$, $V(t) = 30 \cos(2\pi \times 10^6 t) \text{ V}$.

DISPLACEMENT & CONDUCTION CURRENT

Conduction *resistive, $\sigma \neq 0$*

$$E = \frac{V(t)}{d} \quad J_c = \sigma E \text{ (A/m}^2\text{)} \quad I_c = J_c \cdot A \text{ (A)}$$

$$I_c = \underbrace{\frac{\sigma A}{d}}_{1/R} V(t) \longleftrightarrow I = \frac{V}{R} \Rightarrow R = \frac{d}{\sigma A}$$

Current proportional to V directly $\rightarrow$ behaves like a resistor.

Displacement *capacitive, changing E*

$$J_d = \varepsilon \frac{\partial E}{\partial t} \text{ (A/m}^2\text{)} \quad I_d = J_d \cdot A \text{ (A)}$$

$$I_d = \underbrace{\frac{\varepsilon_r \varepsilon_0 A}{d}}_C \frac{dV}{dt} \longleftrightarrow I = C \frac{dV}{dt} \Rightarrow C = \frac{\varepsilon_r \varepsilon_0 A}{d}$$

Current proportional to dV/dt $\rightarrow$ behaves like a capacitor.

Lossy cap circuit *both σ and ε_r present*

$$C = \frac{\varepsilon_r \varepsilon_0 A}{d} = \frac{4 \times 8.85 \times 10^{-12} \times 10^{-3}}{0.02} = \frac{35.4 \times 10^{-15}}{0.02} = 1.77 \times 10^{-12} \text{ F} = 1.77 \text{ pF}$$

$$\frac{dV}{dt} = -30(2\pi \times 10^6) \sin(2\pi \times 10^6 t) = -6\pi \times 10^7 \sin(2\pi \times 10^6 t)$$

$$I_d = C \frac{dV}{dt} = 1.77 \times 10^{-12} \times (-6\pi \times 10^7) \sin(2\pi \times 10^6 t)$$

$$I_d \approx -333 \mu\text{A} \cdot \sin(2\pi \times 10^6 t)$$

Peak magnitude: $|I_d|_{\text{peak}} \approx 333 \mu\text{A}$.

Problem 6 — Lossy Parallel-Plate Capacitor

24 pts

Setup: Plates area A , separation d , material (ϵ_r, σ) , voltage $V(t)$.

DISPLACEMENT & CONDUCTION CURRENT

Conduction *resistive, $\sigma \neq 0$*

$$E = \frac{V(t)}{d} \quad J_c = \sigma E \text{ (A/m}^2\text{)} \quad I_c = J_c \cdot A \text{ (A)}$$

$$I_c = \underbrace{\frac{\sigma A}{d}}_{1/R} V(t) \longleftrightarrow I = \frac{V}{R} \Rightarrow R = \frac{d}{\sigma A}$$

Current proportional to V directly $\rightarrow$ behaves like a resistor.

Displacement *capacitive, changing E*

$$J_d = \epsilon \frac{\partial E}{\partial t} \text{ (A/m}^2\text{)} \quad I_d = J_d \cdot A \text{ (A)}$$

$$I_d = \underbrace{\frac{\epsilon_r \epsilon_0 A}{d}}_C \frac{dV}{dt} \longleftrightarrow I = C \frac{dV}{dt} \Rightarrow C = \frac{\epsilon_r \epsilon_0 A}{d}$$

Current proportional to $dV/dt \rightarrow$ behaves like a capacitor.

Lossy cap circuit *both σ and ϵ_r present*

(a) Conduction current I_c :

$$E = \frac{V(t)}{d}, \quad J_c = \sigma E = \frac{\sigma V(t)}{d} \text{ (A/m}^2\text{)}, \quad \boxed{I_c = J_c \cdot A = \frac{\sigma A}{d} V(t) \text{ (A)}}$$

(b) Displacement current I_d :

$$J_d = \epsilon \frac{\partial E}{\partial t} = \frac{\epsilon_r \epsilon_0}{d} \frac{dV}{dt} \text{ (A/m}^2\text{)}, \quad \boxed{I_d = J_d \cdot A = \frac{\epsilon_r \epsilon_0 A}{d} \frac{dV}{dt} = C \frac{dV}{dt} \text{ (A)}}$$

(c) Equivalent circuit:

From the cheat sheet: $I_c = V/R$ (resistor) and $I_d = C dV/dt$ (capacitor), so the circuit is **R in parallel with C** across $V(t)$, where:

$$R = \frac{d}{\sigma A} \text{ (}\Omega\text{)} \quad C = \frac{\epsilon_r \epsilon_0 A}{d} \text{ (F)}$$

(d) **Evaluate:** $A = 4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$, $d = 0.5 \text{ cm} = 5 \times 10^{-3} \text{ m}$, $\epsilon_r = 4$, $\sigma = 2.5 \text{ S/m}$.

$$R = \frac{5 \times 10^{-3}}{2.5 \times 4 \times 10^{-4}} = \frac{5 \times 10^{-3}}{10^{-3}} = \boxed{5 \Omega}$$

$$C = \frac{4 \times 8.85 \times 10^{-12} \times 4 \times 10^{-4}}{5 \times 10^{-3}} = \frac{14.16 \times 10^{-15}}{5 \times 10^{-3}} = \boxed{2.83 \times 10^{-12} \text{ F} = 2.83 \text{ pF}}$$

Problem 7 — Find Conductivity from Charge Decay

6 pts

Setup: $\epsilon_r = 9$; at $t = 1 \mu\text{s}$, $\rho_v = 10^{-3} \rho_{v0}$.

$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$ Conductor (steady): $\nabla \cdot \mathbf{J} = 0 \Rightarrow \oint_S \mathbf{J} \cdot d\mathbf{s} = 0$ (KCL)		
CHARGE RELAXATION		
Decay charge injected at $t = 0$ $\rho_v(t) = \rho_{v0} e^{-t/\tau_r} \text{ (C/m}^3\text{)} \quad \tau_r = \frac{\epsilon_r \epsilon_0}{\sigma} \text{ (s)}$		

Using the cheat sheet formula $\sigma = -\ln(\rho_v/\rho_{v0}) \cdot \epsilon_r \epsilon_0 / t_1$:

$$\begin{aligned} \sigma &= \frac{-\ln(10^{-3}) \cdot \epsilon_r \epsilon_0}{t_1} = \frac{-(-3 \ln 10) \times 9 \times 8.85 \times 10^{-12}}{10^{-6}} = \frac{3 \times 2.303 \times 9 \times 8.85 \times 10^{-12}}{10^{-6}} \\ &= \frac{6.908 \times 79.65 \times 10^{-12}}{10^{-6}} = \frac{550 \times 10^{-12}}{10^{-6}} \end{aligned}$$

$$\sigma \approx 550 \times 10^{-6} \text{ S/m} = 550 \mu\text{S/m}$$

Problem 8 — Better Insulator

6 pts

Setup: Dry soil: $\epsilon_r = 2.5$, $\sigma = 10^{-4} \text{ S/m}$. Fresh water: $\epsilon_r = 80$, $\sigma = 10^{-3} \text{ S/m}$.

$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$ Conductor (steady): $\nabla \cdot \mathbf{J} = 0 \Rightarrow \oint_S \mathbf{J} \cdot d\mathbf{s} = 0$ (KCL)		
CHARGE RELAXATION		
Decay charge injected at $t = 0$ $\rho_v(t) = \rho_{v0} e^{-t/\tau_r} \text{ (C/m}^3\text{)} \quad \tau_r = \frac{\epsilon_r \epsilon_0}{\sigma} \text{ (s)}$		

From the cheat sheet: larger τ_r = slower dissipation = better insulator.

$$\tau_r^{\text{soil}} = \frac{2.5 \times 8.85 \times 10^{-12}}{10^{-4}} = 221 \times 10^{-9} \text{ s} = 221 \text{ ns}$$

$$\tau_r^{\text{water}} = \frac{80 \times 8.85 \times 10^{-12}}{10^{-3}} = 708 \times 10^{-9} \text{ s} = 708 \text{ ns}$$

$\tau_r^{\text{water}} > \tau_r^{\text{soil}}$, so:

Fresh water is the better insulator ($\tau_r = 708 \text{ ns}$ vs. 221 ns for dry soil)

Problem 9 — Find H from Two-Component E

6 pts

Setup: $\mathbf{E}(z, t) = [4 \cos(6 \times 10^8 t - 2z) \hat{x} + 3 \sin(6 \times 10^8 t - 2z) \hat{y}]$ V/m in air.

Phase	in phase	H lags E by θ_η
<i>Lossless shortcuts (wave in +z, E along $\hat{e}$):</i>		
$\mathbf{E}(z, t)$	$E_0 \cos(\omega t - kz + \phi) \hat{e}$	
$\mathbf{H}(z, t)$	$\frac{E_0}{\eta} \cos(\omega t - kz + \phi) (\hat{k} \times \hat{e})$	
$\mathbf{H}(t)$ direct	$\frac{1}{\eta} (\hat{k} \times \mathbf{E}(t))$	
$\mathbf{E}(t)$ direct	$-\eta (\hat{k} \times \mathbf{H}(t))$	

DIRECTION OF PROPAGATION

Order: $\omega \rightarrow u_p \rightarrow k \rightarrow \lambda \rightarrow \eta$. Base: $\mathbf{E} = E_0 \cos(\omega t \pm kx + \phi_0) \hat{e}$.

LOSSLESS PLANE WAVE — E AND H

General E

$\mathbf{E} = E_0 \cos(\omega t - k \hat{k} \cdot \mathbf{r} + \phi_0) \hat{e}$

$\hat{k}$ = direction of travel (unit vector), e.g. $+y \Rightarrow \hat{k} = \hat{y}$.
 k = wavenumber (rad/m): how fast phase changes in space;
 $k = \omega \sqrt{\epsilon_r} / c = 2\pi / \lambda$.
 ϕ_0 = initial phase constant (rad): shift set by given condition at known t , position.
 $\hat{e}$ = polarization direction of $\mathbf{E}$ (unit vector), e.g. E along $x \Rightarrow \hat{e} = \hat{x}$.
 $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, so $\hat{k} \cdot \mathbf{r}$ picks the axis, e.g. $\hat{k} = \hat{y} \Rightarrow \hat{k} \cdot \mathbf{r} = y$.

Arg. contains	$\hat{k}$	Arg. contains	$\hat{k}$
$\omega t - kz$	$+\hat{z}$	$\omega t + kz$	$-\hat{z}$
$\omega t - ky$	$+\hat{y}$	$\omega t + ky$	$-\hat{y}$
$\omega t - kx$	$+\hat{x}$	$\omega t + kx$	$-\hat{x}$

Minus before k -axis $\Rightarrow$ + direction. Plus $\Rightarrow$ - direction.

$\hat{k} \times \hat{E}$ — FIND $\hat{H}$ DIRECTION

$\hat{k}$	$\hat{E}$	$\hat{H}$	$\hat{k}$	$\hat{E}$	$\hat{H}$
$+\hat{z}$	$\hat{x}$	$+\hat{y}$	$+\hat{z}$	$\hat{y}$	$-\hat{x}$
$-\hat{z}$	$\hat{x}$	$-\hat{y}$	$-\hat{z}$	$\hat{y}$	$+\hat{x}$
$+\hat{y}$	$\hat{x}$	$-\hat{z}$	$+\hat{y}$	$\hat{z}$	$+\hat{x}$
$-\hat{y}$	$\hat{x}$	$+\hat{z}$	$-\hat{y}$	$\hat{z}$	$-\hat{x}$

Arg contains $-kz \Rightarrow \hat{k} = +\hat{z}$. Air: $\eta = \eta_0 = 377 \Omega$.

Note: $3 \sin(\omega t - kz) = 3 \cos(\omega t - kz - \pi/2)$ — same envelope, just phase-shifted.

Use $\mathbf{H} = \frac{1}{\eta} (\hat{k} \times \mathbf{E})$:

$$\begin{aligned}
 \hat{k} \times \mathbf{E} &= \hat{z} \times [4 \cos(\omega t - kz) \hat{x} + 3 \sin(\omega t - kz) \hat{y}] \\
 &= 4 \cos(\omega t - kz) \underbrace{(\hat{z} \times \hat{x})}_{\hat{y}} + 3 \sin(\omega t - kz) \underbrace{(\hat{z} \times \hat{y})}_{-\hat{x}} \\
 &= 4 \cos(\omega t - kz) \hat{y} - 3 \sin(\omega t - kz) \hat{x}
 \end{aligned}$$

$$\mathbf{H}(z, t) = \frac{1}{377} [-3 \sin(6 \times 10^8 t - 2z) \hat{x} + 4 \cos(6 \times 10^8 t - 2z) \hat{y}] \text{ A/m}$$

$$\approx [-7.96 \times 10^{-3} \sin(6 \times 10^8 t - 2z) \hat{x} + 10.6 \times 10^{-3} \cos(6 \times 10^8 t - 2z) \hat{y}] \text{ A/m}$$

Problem 10 — Find $\mathbf{H}$ Using Faraday (Non-Uniform $\mathbf{E}$)

8 pts

Setup: $\mathbf{E} = E_0 \sin(ay) \cos(\omega t - kz) \hat{x}$.

Cannot use $\mathbf{H} = \frac{1}{\eta}(\hat{k} \times \mathbf{E})$ here because $\mathbf{E}$ varies in y (not a pure plane wave). Must use Faraday's law with the full curl.

Gauss (B) $\nabla \cdot \mathbf{B} = 0$ $\mathbf{D} = \epsilon \mathbf{E}; \mathbf{B} = \mu \mathbf{H};$ source-free: $\rho_v = 0, \mathbf{J} = \mathbf{0}$ SURFACE ELEMENT ds $ds = \hat{n} da$ (RHR gives $\hat{n}$: curl fingers in loop direction, thumb = $\hat{n}$) <table border="1"> <tr> <td>Loop plane</td> <td>ds</td> </tr> </table>	Loop plane	ds	$\rho_v (C/m^3); \sigma (S/m); \epsilon_0 = 8.85 \times 10^{-12} F/m$ Find σ from decay data $\sigma = \frac{-\ln(\rho_v / \rho_{v0}) \cdot \epsilon_r \epsilon_0}{t_1}$ larger τ_r = slower dissipation = better insulator. PHASOR METHOD ($\mathbf{E} \rightarrow \mathbf{H}$) Key relation $\mathbf{E}(z, t) = \text{Re}\{\tilde{\mathbf{E}}(z) e^{j\omega t}\}$ Euler split $e^{j(\omega t - kz)} = e^{j\omega t} \cdot e^{-jkz}$ strip $e^{j\omega t}$, keep e^{-jkz}
Loop plane	ds		

Apply $\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t}$. With $E_x = E_0 \sin(ay) \cos(\omega t - kz)$, $E_y = E_z = 0$:

$$\nabla \times \mathbf{E} = -\hat{y} \frac{\partial E_x}{\partial z} + \hat{z} \frac{\partial E_x}{\partial y}$$

$$\frac{\partial E_x}{\partial z} = E_0 \sin(ay) \cdot k \sin(\omega t - kz)$$

$$\frac{\partial E_x}{\partial y} = E_0 \cdot a \cos(ay) \cos(\omega t - kz)$$

$$-\mu_0 \frac{\partial \mathbf{H}}{\partial t} = -\hat{y} k E_0 \sin(ay) \sin(\omega t - kz) + \hat{z} a E_0 \cos(ay) \cos(\omega t - kz)$$

Integrate over t ($\int \sin(\omega t - kz) dt = -\frac{1}{\omega} \cos(\omega t - kz)$; $\int \cos(\omega t - kz) dt = \frac{1}{\omega} \sin(\omega t - kz)$):

$$\mathbf{H} = \frac{1}{\mu_0} \left[\hat{y} \frac{k}{\omega} E_0 \sin(ay) \cdot \frac{-\cos(\omega t - kz)}{1} \cdot \frac{1}{1} - \hat{z} \frac{a}{\omega} E_0 \cos(ay) \sin(\omega t - kz) \right]$$

$$\mathbf{H}(z, t) = -\frac{k E_0}{\mu_0 \omega} \sin(ay) \cos(\omega t - kz) \hat{y} - \frac{a E_0}{\mu_0 \omega} \cos(ay) \sin(\omega t - kz) \hat{z}$$

All answers derived using ECE332 Exam 1 Cheat Sheet (HW1+HW2 pages)